

CALCULUS II

WEEK VIII

PARTIAL DERIVATIVES

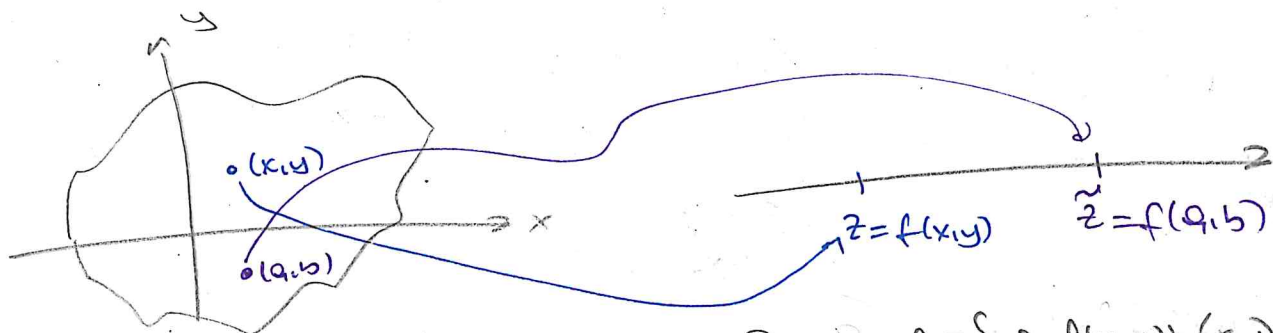
Definition A function of n real variables is a rule that assigns a unique real number $f(x_1, x_2, \dots, x_n)$ to each point $(x_1, x_2, \dots, x_n)$ in some set $D(f)$ of $\mathbb{R}^n$. $D(f)$ is called the domain of f .

The set of real numbers $f(x_1, x_2, \dots, x_n)$ obtained from points in the domain is called the range of f .

→ As for functions of one variable, the domain convention specifies that the domain of a function of n variables is the largest set of points $(x_1, x_2, \dots, x_n)$ for which $f(x_1, x_2, \dots, x_n)$ makes sense as a real number, unless that domain is explicitly stated to be a smaller set.

→ If f is a function of two independent variables, we call these independent variables x and y and picture the domain of f as a region in the xy -plane.

We often write $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ z : dependent variable
 $(x, y) \mapsto z = f(x, y)$ x, y : independent variables

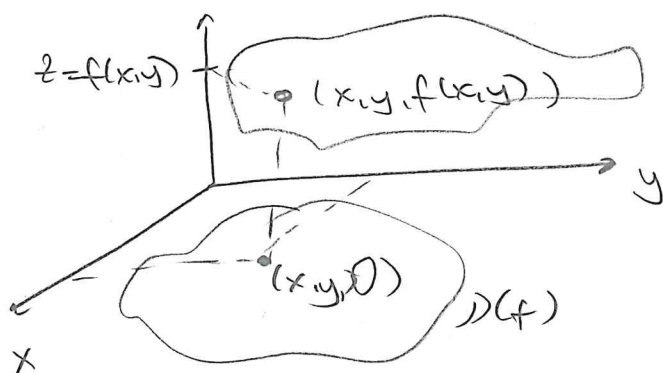


Domain of f $D(f)$

Range of $f = \{ z = f(x, y) : (x, y) \in D(f) \}$

GRAPHS

→ If f is a function of two variables with domain $D(f)$, then the graph of f is the set of all points $(x, y, z) \in \mathbb{R}^3$ such that $z = f(x, y)$



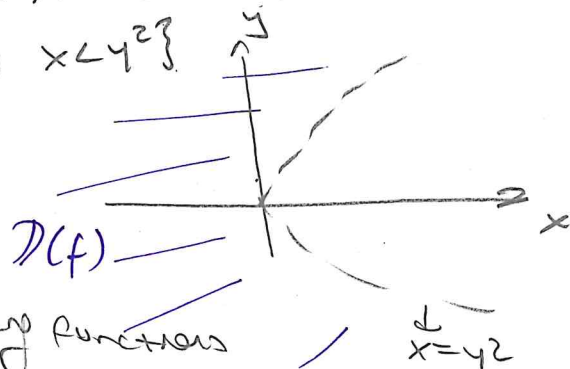
The graph of $f(x, y)$ is the surface with equation $z = f(x, y)$ defined for points (x, y) in the domain of f

Ex 7 Find domain of $f(x, y) = x \ln(y^2 - x)$ and evaluate $f(1, 2)$

- $f(1, 2) = 1 \cdot \ln(4 - 1) = \ln 3$

- $\ln(y^2 - x)$ only defined when $y^2 - x > 0 \Rightarrow x < y^2$

The domain of f $D(f) = \{(x, y) \mid x < y^2\}$



Ex 7 Find domain and range of the following functions

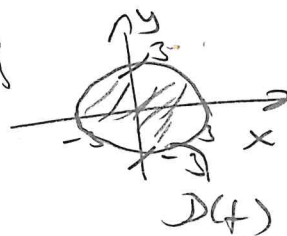
① $z = f(x, y) = \sqrt{9 - x^2 - y^2}$

$D(f) = \{(x, y) \mid 9 - x^2 - y^2 \geq 0\} = \{(x, y) \mid x^2 + y^2 \leq 9\}$

$R(f) = \{z \mid z = \sqrt{9 - x^2 - y^2} \text{ } (x, y) \in D(f)\}$

z is a positive square root so $z \geq 0$ also

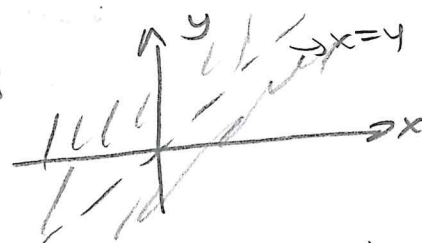
$9 - x^2 - y^2 \leq 9 \Rightarrow \sqrt{9 - x^2 - y^2} \leq 3$ So the range $R(f) = \{z \mid 0 \leq z \leq 3\} = [0, 3]$



② $f(x, y) = \frac{x+y}{x-y}$

$D(f) = \{(x, y) \mid x \neq y\}$

$R(f) = (-\infty, \infty)$



All cartesian coordinate system except

③ $f(x,y) = \frac{1}{xy}$

$D(f) = \{(x,y) : xy \neq 0\}$
 $R(f) = (-\infty, \infty) \setminus \{0\}$



All cartesian coordinate
 except $x=0 \Rightarrow y$ -axis
 $y=0 \Rightarrow x$ -axis.

④ $f(x,y) = \sqrt{9-x^2} - \sqrt{4-y^2}$

We must have $9-x^2 \geq 0$ and $4-y^2 \geq 0$. Then, we get the

domain $D(f) = \{(x,y) \in \mathbb{R}^2 : -3 \leq x \leq 3, -2 \leq y \leq 2\}$

$f(x,y) \leq \sqrt{9-x^2}$ $\sqrt{9-x^2}$ takes its max at $x=0$ so $f(x,y) \leq 3$

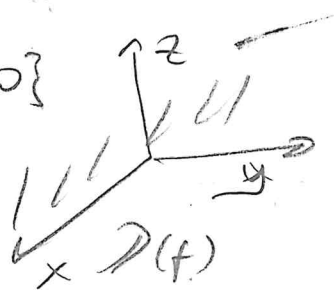
Similarly, $f(x,y) \geq -\sqrt{4-y^2}$ its max is at $x=0$ which gives $f(x,y) \geq -2$

So, the range $R(f) = [-2, 3]$

⑤ $f(x,y,z) = xyz$ (will be solved in 3-dim)

$D(f) = \{(x,y,z) : x \in (-\infty, \infty), y \in (-\infty, \infty), z > 0\}$

$R(f) = (-\infty, \infty)$



⑥ $f(x,y,z) = \frac{x+y+z}{|x+y+z|}$ (will be solved in 3-dim)

$D(f) = \{(x,y,z) \in \mathbb{R}^3 : x+y+z \neq 0\}$

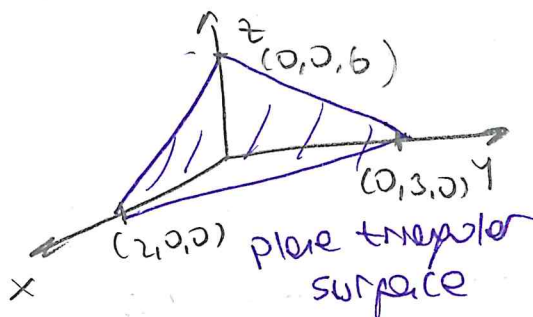
$R(f) = \{-1, 1\}$

Ex 7 Sketch the graph of the function $f(x,y) = 6 - 3x - 2y$
 $z = 6 - 3x - 2y$ represents a plane

For $y=z=0 \Rightarrow x=2$

For $x=z=0 \Rightarrow y=3$

For $x=y=0 \Rightarrow z=6$

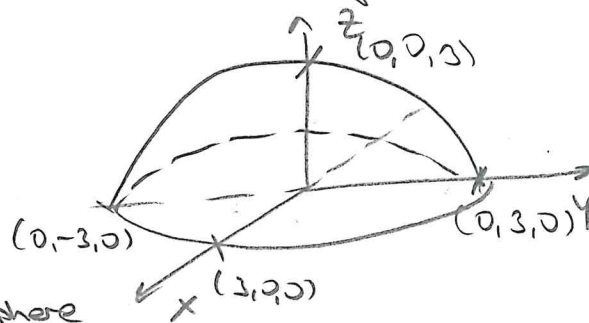


$f(x,y) = ax+by+c$ $a,b,c \in \mathbb{R}$ (linear function)

Ex 7 Sketch the graph of the function $f(x,y) = \sqrt{9-x^2-y^2}$

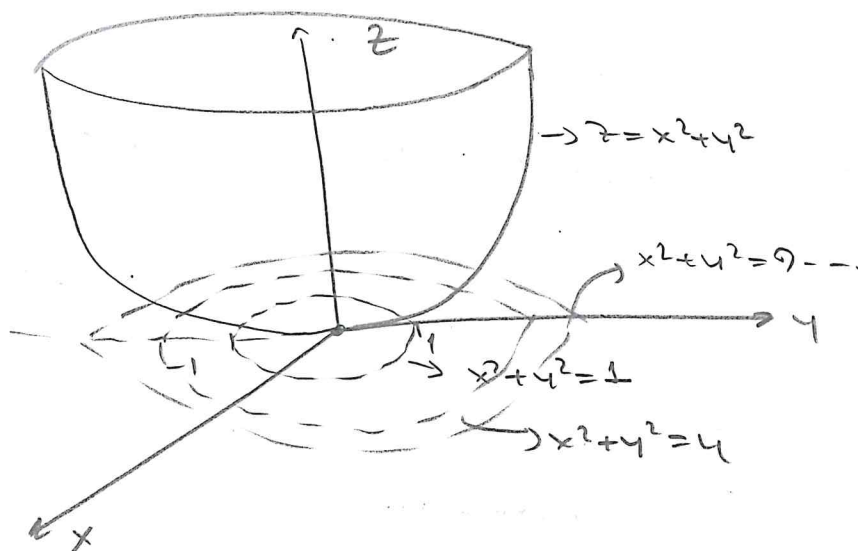
$z = \sqrt{9-x^2-y^2} \Rightarrow x^2+y^2+z^2=9$ the equation of the sphere centered at the origin with radius 3

But note $z \geq 0$ hence, the graph of $f(x,y)$ is the top half of the sphere



LEVEL CURVES

→ In the xy -plane we sketch the curves $f(x,y)=C$ for various values of the constant C . These curves are called level curves of f because they are the vertical projections onto the xy -plane of the curves in which the graph $z=f(x,y)$ intersects the horizontal (level) planes $z=C$



Ex 7 Sketch the level curves of the function $f(x,y) = 6-3x-2y$

$C = -6, 0, 6, 12 \quad f(x,y) = C$

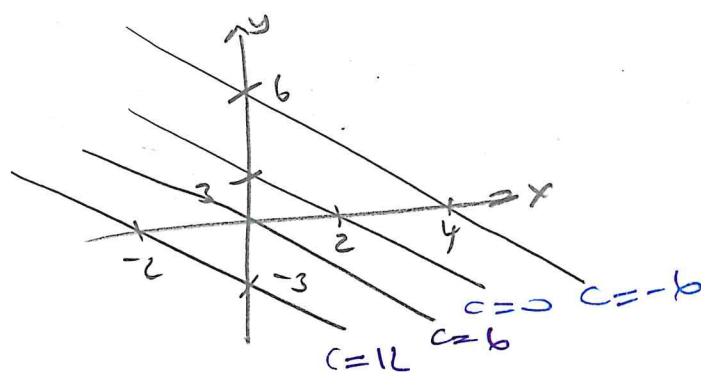
$6-3x-2y = -6 \Rightarrow 3x+2y=12$

$6-3x-2y = 0 \Rightarrow 3x+2y=6$

$6-3x-2y = 6 \Rightarrow 3x+2y=0$

$6-3x-2y = 12 \Rightarrow 3x+2y=-6$

Family of lines with slope $-3/2$

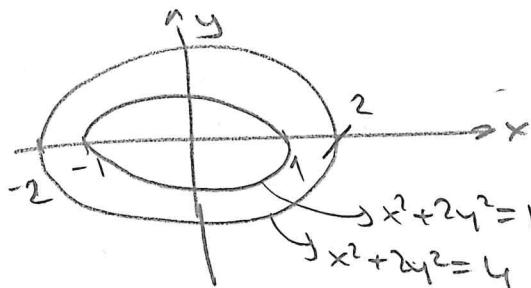


Ex 7 Sketch the level curves of $f(x,y) = x^2 + 2y^2$

$$x^2 + 2y^2 = C$$

$$C=1 \Rightarrow x^2 + 2y^2 = 1$$

$$C=4 \Rightarrow x^2 + 2y^2 = 4$$



Level surfaces Level surfaces are the surfaces with equations

$$f(x,y,z) = C, \quad C \text{ is a constant}$$

Ex 7 Find the level surfaces of the function $f(x,y,z) = x^2 + y^2 + z^2$

$x^2 + y^2 + z^2 = C \quad C > 0$. These form a family of spheres centered at the origin with radius $\sqrt{C}$

LIMITS AND CONTINUITY

Definition of Limit

We say that $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$, provided that

① every neighbourhood of (a,b) contains points of the domain of f different from (a,b) , and

② for every positive number ϵ there exists a positive number $\delta = \delta(\epsilon)$ such that $|f(x,y) - L| < \epsilon$ holds whenever (x,y) is in the domain of f and satisfies

$$0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$$

Formal Definition

For every $\epsilon > 0$ there is a corresponding $\delta > 0$ such that

$|f(x,y) - L| < \epsilon$ whenever $(x,y) \in D(f)$ and $0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta$

distance between the point (x,y) and the point (a,b) , circle center $M(a,b)$ radius δ

distance between real numbers $f(x,y)$ and L absolute value.

→ All the usual laws of limits extend to functions of several variables in the obvious way.

$$\text{Let } \lim_{(x,y) \rightarrow (a,b)} f(x,y) = L \text{ and } \lim_{(x,y) \rightarrow (a,b)} g(x,y) = M \text{ and}$$

every neighbourhood of (a,b) contains points in

$D(f) \cap D(g)$ other than (a,b) , then

$$\lim_{(x,y) \rightarrow (a,b)} (f(x,y) \mp g(x,y)) = L \mp M$$

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) \cdot g(x,y) = L \cdot M$$

$$\lim_{(x,y) \rightarrow (a,b)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M} \text{ provided } M \neq 0$$

Also, If $F(t)$ is continuous at $t=L$, then

$$\lim_{(x,y) \rightarrow (a,b)} F(f(x,y)) = F(L)$$

Remark The limit definition says that the distance between $f(x,y)$ and L can be arbitrary small by making the distance (x,y) to (a,b) sufficiently small. The definition refers only to the distance between (x,y) and (a,b) . It DOES NOT refer to the direction of approach. Therefore, if

the limit exists, then $f(x,y)$ must approach the

Same limit no. matter how (x,y) approaches to (a,b)

* If $f(x,y) \rightarrow L_1$ as $(x,y) \rightarrow (a,b)$ along path C_1 ,
 and $f(x,y) \rightarrow L_2$ as $(x,y) \rightarrow (a,b)$ along path C_2 ,
 where $L_1 \neq L_2$, then $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ doesn't exist.

Ex 1 Evaluate the following limits?

a) $\lim_{(x,y) \rightarrow (0,2)} \frac{x - xy + 3}{x^2y - y^2} = ?$

$$\lim_{(x,y) \rightarrow (0,2)} \frac{x - xy + 3}{x^2y - y^2} = \frac{0 - 0 + 3}{0 - 4} = -3/4$$

b) $\lim_{(x,y) \rightarrow (2,-1)} xy + x^2 = -2 + 4 = 2$

c) $\lim_{(x,y) \rightarrow (0,0)} \sqrt{x^2 + y^2} = 0$

d) $\lim_{(x,y) \rightarrow (2,3)} (2x - y^2) = 4 - 9 = -5$

e) $\lim_{(x,y) \rightarrow (0,b)} x^2y = 0^2b = 0$

f) $\lim_{(x,y) \rightarrow (\pi/3, 2)} y \cdot \sin(x/y) = 2 \sin(\pi/6) = 1$

g) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} = ?$

$$f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$$

→ Let's approach $(0,0)$ along the x-axis. Then, $y=0$ this
 gives $f(x,0) = \frac{x^2}{x^2} = 1$ for all $x \neq 0$. So, $f(x,y) \rightarrow 1$ as $(x,y) \rightarrow (0,0)$
 along x-axis

→ Let's approach $(0,0)$ along the y-axis. Then, $x=0$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{-y^2}{y^2} = -1. \quad f(x,y) \rightarrow -1 \text{ as } (x,y) \rightarrow (0,0) \text{ along y-axis}$$

Since, f has two different limits we say

LIMIT DOES NOT EXIST

(h) $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2+y^2} = ?$

Approach along x-axis: $f(x,0) = \frac{0}{x^2} = 0 \quad x \neq 0$

Approach along y-axis: $f(0,y) = \frac{0}{y^2} = 0 \quad y \neq 0$

We have identical limits along the axes. BUT that doesn't show that the given limit is 0.

Approach along $y=x$: $f(x,x) = \frac{x^2}{2x^2} = \frac{1}{2}$

Since, we have obtained different limits along different paths the given limit doesn't exist.

(I) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - xy}{\sqrt{x} - \sqrt{y}} = ?$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x(x-y)}{\sqrt{x} - \sqrt{y}} = \lim_{(x,y) \rightarrow (0,0)} \frac{x(\sqrt{x} - \sqrt{y})(\sqrt{x} + \sqrt{y})}{(\sqrt{x} - \sqrt{y})} = 0$$

(J) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 + y^2}{y} = ?$

Along y axis: $f(0,y) = \frac{y^2}{y} = y \quad \lim_{(x,y) \rightarrow (0,0)} y = 0$

Along $y=x^2$: $f(x,x^2) = \frac{x^2 + x^4}{x^2} = 1 + x^2 \quad \lim_{(x,y) \rightarrow (0,0)} 1 + x^2 = 1$

$1 \neq 0$ LIMIT DOESN'T EXIST

(K) $\lim_{(x,y) \rightarrow (0,0)} \frac{2xy}{x^2+y^2} = ?$

Along x-axis: ($y=0$): $f(x,0) = \frac{0}{x^2} \quad \lim_{(x,y) \rightarrow (0,0)} f(x,0) = 0$

Along $y=x$: $f(x,x) = \frac{2x^2}{2x^2} = 1 \quad \lim_{(x,y) \rightarrow (0,0)} f(x,x) = 1$

$1 \neq 0$ LIMIT DOESN'T EXIST

II WAY $y=kx$ $f(x,y) = \frac{2xy}{x^2+y^2}$ $f(x,kx) = \frac{2kx^2}{x^2+k^2x^2} = \frac{2k}{1+k^2}$

For each $k \neq 0$ we obtain different values. Therefore, limit doesn't exist

Ex 2 Investigate the limiting behavior of $f(x,y) = \frac{2x^2y}{x^4+y^2}$ as (x,y) approaches $(0,0)$

$y=kx$ $f(x,kx) = \frac{2kx^3}{x^4+k^2x^2} = \frac{2kx}{x^2+k^2}$

$\lim_{(x,y) \rightarrow (0,0)} f(x,kx) = 0$

Consider $y=x^2$ $f(x,x^2) = \frac{2x^4}{x^4+x^4} = 1$

Therefore, limit doesn't exist.

Ex 3 Show that the function $f(x,y) = \frac{x^2y}{x^2+y^2}$ doesn't have a limit at the origin. Specifically,

$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y}{x^2+y^2} = 0$

bl This, function is defined everywhere except at the origin. Observe that, since $x^2 \leq x^2+y^2$, we have $(\frac{1}{x^2+y^2} \leq \frac{1}{x^2})$

$|f(x,y) - 0| = \left| \frac{x^2y}{x^2+y^2} \right| \leq \left| \frac{x^2y}{x^2} \right| \leq |y| \leq \sqrt{x^2+y^2}$
 $x^2 > 0$

which approaches zero as $(x,y) \rightarrow (0,0)$

Formally, if $\epsilon > 0$ is given we take $\delta = \epsilon$. Then,

$|f(x,y) - 0| < \epsilon$ whenever $\sqrt{x^2+y^2} < \delta$

$|f(x,y) - L| < \epsilon$ whenever $(9) \sqrt{(x-a)^2 + (y-b)^2} < \delta$

SQUEEZE (SANDWICH) THEOREM

For functions of two variables states that if

$$g(x,y) \leq f(x,y) \leq h(x,y) \text{ for all } (x,y) \neq (a,b)$$

$$\text{and if } \lim_{(x,y) \rightarrow (a,b)} g(x,y) = L \quad \lim_{(x,y) \rightarrow (a,b)} h(x,y) = L$$

$$\text{Then, we get } \lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$$

$$\text{Ex 4) } \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2} = ?$$

$$\text{we have } \frac{x^2}{x^2 + 2y^2} \leq 1$$

$$0 \leq \frac{x^2 \sin^2 y}{x^2 + 2y^2} \leq \sin^2 y$$

$$\lim_{(x,y) \rightarrow (0,0)} 0 = 0 \quad \lim_{(x,y) \rightarrow (0,0)} \sin^2 y = 0$$

$$\text{Using squeeze theorem gives } \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 \sin^2 y}{x^2 + 2y^2} = 0$$

$$\text{Ex 5) } \lim_{(x,y,z) \rightarrow (0,0,0)} \frac{x^2 + 2y^2 + 3z^2}{x^2 + y^2 + z^2} = ? \quad (\text{Not a sandwich theorem question})$$

$$\text{Along y axis } f(0,y,0) = \frac{2y^2}{y^2} = 2$$

$$\text{Along x axis } f(x,0,0) = \frac{x^2}{x^2} = 1$$

$$\lim_{(x,y,z) \rightarrow (0,0,0)} f(x,0,0) \neq \lim_{(x,y,z) \rightarrow (0,0,0)} f(0,y,0) \neq \lim_{(x,y,z) \rightarrow (0,0,0)} f(x,0,z)$$

LIMIT DOESN'T EXIST

CONTINUITY

A function $f(x,y)$ is called continuous at (a,b) if

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

We say that f is continuous on D if f is continuous at every point in (a,b) in D .

NOTE (1) All polynomials are continuous on $\mathbb{R}^2$

(2) Any rational function $\left(\frac{\text{polynomial}}{\text{polynomial}} \right)$ is continuous on its domain,

(3) Sums, differences, products, quotients and composition of continuous functions are continuous

Ex 6 $\lim_{(x,y) \rightarrow (1,2)} (x^2y^3 - x^3y^2 + 3x + 2y) = 11$

$f(x,y) = x^2y^3 - x^3y^2 + 3x + 2y$ is a polynomial, continuous everywhere, the limit can be found by direct substitution.

Ex 7 Where is the function $f(x,y) = \frac{x^2 - y^2}{x^2 + y^2}$ continuous?

The function is not continuous at $(0,0)$, because it is not defined at $(0,0)$. $f(x,y)$ is continuous at its domain that is $D(f) = \{(x,y) : (x,y) \neq (0,0)\}$

Ex 8 $h(x,y) = \begin{cases} \frac{x^2 - y^2}{x^2 + y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ $h(x,y)$ is defined at $(0,0)$ but it is not continuous.

Since we proved that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$ doesn't exist

Proved Ex (1) (8)

Ex 9 $f(x,y) = \frac{xy}{x+y}$ is a rational function. It is not defined for $y = -x$.

for $y = -x$. It is continuous on its domain $D(f) = \{(x,y) \mid y \neq -x\}$.

Ex 10 $\lim_{(x,y,z) \rightarrow (2,1,-1)} 3x^2z + y \cos(\pi x - \pi z) = 3 \cdot 4(-1) + 1 \cdot 2 \cos(2\pi + \pi) = -14$

x^2z : polynomial continuous. $\cos(\pi x - \pi z)$: continuous

y : polynomial continuous

Ex 11 Show that $f(x,y) = \begin{cases} \frac{2xy}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 0 & \text{if } (x,y) = (0,0) \end{cases}$

continuous at every point except the origin

What about at $(x,y) = (0,0)$?

Approach along $y = mx$ $f(x, mx) = \frac{2mx^2}{x^2 + m^2x^2} = \frac{2m}{1+m^2}$

Hence, $f(x,y) \rightarrow \frac{2m}{1+m^2}$ as $(x,y) \rightarrow (0,0)$ along $y = mx$

Thus, the limit changes with m . Therefore, there is no single number for the limit of $f(x,y)$ as $(x,y) \rightarrow (0,0)$.

Hence, function is not continuous at $(0,0)$

Ex 12 $f(x,y) = \begin{cases} \frac{(x-2)(y-1)}{(x-2)^2 + (y-1)^2} & (x,y) \neq (2,1) \\ 0 & (x,y) = (2,1) \end{cases}$

$\lim_{(x,y) \rightarrow (2,1)} \frac{(x-2)(y-1)}{(x-2)^2 + (y-1)^2} \Rightarrow$ Approach along $x = 2y$ $\lim_{(x,y) \rightarrow (2,1)} \frac{(x-2)(y-1)}{(x-2)^2 + (y-1)^2} = \lim_{(2y,y) \rightarrow (2,1)} \frac{(2y-2)(y-1)}{(2y-2)^2 + (y-1)^2} = \frac{2(y-1)}{5(y-1)^2} = \frac{2}{5}$

Along $x = y+1$ $\lim_{(x,y) \rightarrow (2,1)} \frac{(x-2)(y-1)}{(x-2)^2 + (y-1)^2} = \lim_{(x,y) \rightarrow (2,1)} \frac{(y-1)(y-1)}{(y-1)^2 + (y-1)^2} = \frac{1}{2}$

Along $x = 2$ $\lim_{x \rightarrow 2} \frac{(x-2)(y-1)}{(x-2)^2 + (y-1)^2} = 0$

$\frac{2}{5} \neq \frac{1}{2} \neq 0$ Limit doesn't exist NOT CONT.

Ex 13 $f(x,y) = \begin{cases} 0 & \text{if } (x,y) \neq (0,0) \\ 1 & \text{if } (x,y) = (0,0) \end{cases}$ cont at $(0,0)$?

$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0 \neq f(0,0) = 1$ NOT CONT.

As for functions of one variable, the existence of a limit of a function at a point, does not imply continuity at that point.

PARTIAL DERIVATIVES

The first partial derivatives of the function $f(x,y)$ with respect to variables x and y are the functions $f_1(x,y)$ and $f_2(x,y)$ given by

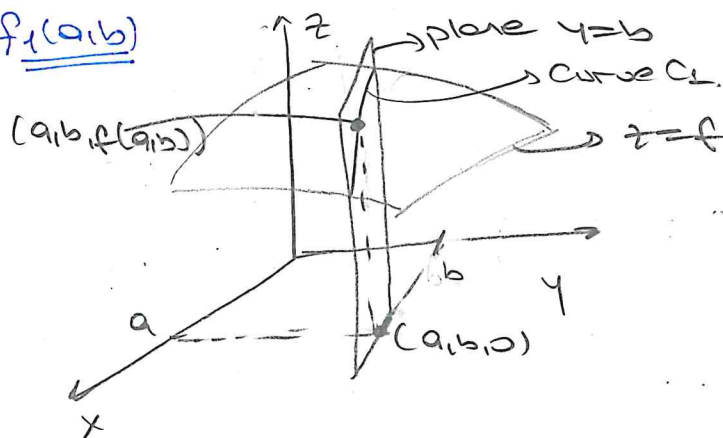
$$f_1(x,y) = \lim_{h \rightarrow 0} \frac{f(x+h,y) - f(x,y)}{h} = f_x(x,y) = \frac{\partial f}{\partial x}$$

$$f_2(x,y) = \lim_{k \rightarrow 0} \frac{f(x,y+k) - f(x,y)}{k} = f_y(x,y) = \frac{\partial f}{\partial y}$$

Ex If $f(x,y) = x^2 \sin y$ $f_x(x,y) = 2x \sin y$ $f_y(x,y) = x^2 \cos y$

Interpretations of Partial Derivatives

$f_1(a,b)$



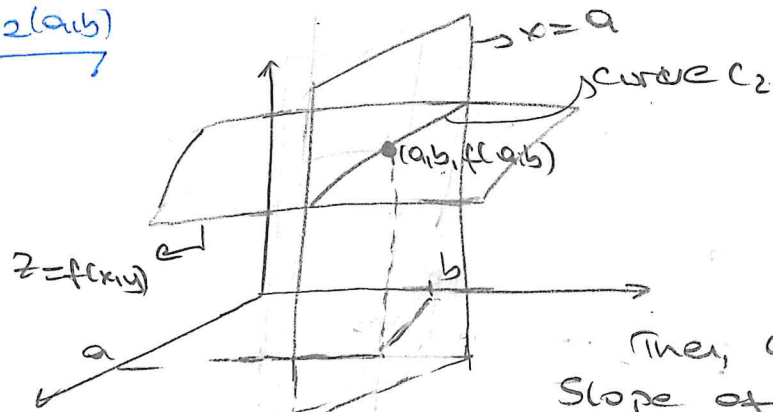
Fix $y=b$ and C_1 is the intersection of $z=f(x,y)$

and $y=b$.

Then $C_1: g(x) = f(x,b)$, so the Slope of this line is $g'(x)$ and at point $(a,b,f(a,b))$ is $g'(a) = f_1(a,b)$

Therefore, $f_1(a,b)$ is the slope of the curve of intersection $z = f(x,y)$ and the vertical plane $y=b$ at $x=a$.

$f_2(a,b)$



→ For $x=a$ then
 C_2 is the curve
of the intersection
 $z=f(x,y)$ and $x=a$.

Then, $C_2 \equiv f(a,y) = h(y)$
Slope of this line is $h'(y)$ and at
point $(a,b,f(a,b))$ is $h'(b)$.

Therefore, $f_2(x,y)$ is the slope of the curve of the intersection
 $z=f(x,y)$ and horizontal curve $x=a$ at $y=b$

Notations for first partial derivatives

$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} f(x,y) = f_1(x,y) = D_1 f(x,y)$$

$$\frac{\partial z}{\partial y} = \frac{\partial}{\partial y} f(x,y) = f_2(x,y) = D_2 f(x,y)$$

Values of partial derivatives

$$\left. \frac{\partial z}{\partial x} \right|_{(a,b)} = \left(\frac{\partial}{\partial x} f(x,y) \right) \Big|_{(a,b)} = f_1(a,b) = D_1 f(a,b)$$

$$\left. \frac{\partial z}{\partial y} \right|_{(a,b)} = \left(\frac{\partial}{\partial y} f(x,y) \right) \Big|_{(a,b)} = f_2(a,b) = D_2 f(a,b)$$

Ex Obtain the following partial derivatives.

(a) $w(x, y, z) = x^2y - 10y^2z^3 + 43x - 7 \tan(4y)$

$$w_x = 2xy + 43 \quad w_y = x^2 - 20yz^3 - 28 \sec^2(4y) \quad w_z = -30y^2z^2$$

(b) $h(s, t) = t^7 \ln(s^2) + \frac{9}{t^3} - 7\sqrt{s^4}$

$$h_t = 7t^6 \ln(s^2) - \frac{27}{t^2} \quad h_s = t^7 \cdot \frac{2s}{s^2} - \frac{4}{7} s^{-3/2}$$

(c) (solve after chain rule given page 17)
 $x^2 \sin(2y - 5z) = 1 + 4 \cos(6zx)$ find z_x, z_y if $z = z(x, y)$

Derivative w.r.t x

$$2x \sin(2y - 5z) - 5x^2 \cos(2y - 5z) z_x = -y(bz + b x z_x) \sin(6zx)$$

$$z_x (bxy \sin(6zx) - 5x^2 \cos(2y - 5z)) = -b y z \sin(6zx) - 2x \sin(2y - 5z)$$

$$z_x = \frac{2x \sin(2y - 5z) + b y z \sin(6zx)}{5x^2 \cos(2y - 5z) - bxy \sin(6zx)}$$

Derivative w.r.t y

$$2x^2 \cos(2y - 5z) + 2x^2 \sin(2y - 5z) \cdot 5z_y$$

$$= -\cos(6zx) - y \sin(6zx) b x z_y$$

$$z_y \{ -5x^2 \cos(2y - 5z) + bxy \sin(6zx) \} = \cos(6zx) - 2x^2 \cos(2y - 5z)$$

$$z_y = \frac{\cos(6zx) - 2x^2 \cos(2y - 5z)}{bxy \sin(6zx) - 5x^2 \cos(2y - 5z)}$$

Ex Obtain the derivatives of the following functions at (0,0) point by using the definition of the derivative

(a) $f(x,y) = \begin{cases} \frac{2x^3 - y^3}{x^2 + 3y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$ (b) $f(x,y) = \begin{cases} \frac{x^2 - 2y^2}{x - y} & x \neq y \\ 0 & x = y \end{cases}$

(a) $\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{\frac{2h^3 - 0}{h^2 + 0} - 0}{h} = \lim_{h \rightarrow 0} \frac{2h^3 - 0}{h^2} = ?$

$\frac{\partial f}{\partial y}(0,0) = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k \rightarrow 0} \frac{\frac{-k^3}{3k^2} - 0}{k} = -1/3$

(b) $\frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{h^2/h - 0}{h} = 1$

$\frac{\partial f}{\partial y}(0,0) = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k \rightarrow 0} \frac{\frac{-2k^2}{-k} - 0}{k} = 2$

Ex $z = x^2y^2 + x^4y + y^4$ $z_x = ?$ $z_y = ?$
 $z_x = 2x^2y^2 + 4x^3y$ $z_y = 2x^3y + x^4 + 4y^3$

Ex $f(x,y) = e^{xy} \cos(x+y)$ $f_x(0,\pi) = ?$

$f_x(x,y) = e^{xy} y \cos(x+y) - \sin(x+y) e^{xy}$

$f_x(0,\pi) = e^0 \pi \cos(\pi) - \sin(\pi) \cdot e^0 = -\pi$

CHAIN RULE

The single-variable version of the Chain Rule also continues to apply to, say $f(g(x,y))$ where f is a function of only one variable having derivative f'

$$\frac{\partial}{\partial x} f(g(x,y)) = f'(g(x,y)) g_x(x,y)$$

$$\frac{\partial}{\partial y} f(g(x,y)) = f'(g(x,y)) \cdot g_y(x,y)$$

Ex Show that $z = f(x/y)$ satisfies the partial differential equation $x \cdot z_x + y \cdot z_y = 0$

$$z_x = f'(x/y) (1/y) \quad z_y = f'(x/y) (-x/y^2)$$

$$x z_x + y z_y = f'(x/y) \left[x \cdot \frac{1}{y} - \frac{x}{y^2} \cdot y \right] = 0$$

Ex Determine if $f(x,y) = \frac{x^2}{y^3}$ is increasing or decreasing at $(2,5)$

(a) If we allow x to vary and hold y fixed

(b) If we allow y to vary and hold x fixed

(a) $f_x(x,y) \Big|_{(2,5)} = \frac{2x}{y^3} \Big|_{(2,5)} = \frac{4}{125} > 0$ the function is increasing at $(2,5)$ as we vary x and hold y fixed

(b) $f_y(x,y) \Big|_{(2,5)} = -\frac{3x^2}{y^4} \Big|_{(2,5)} = -\frac{12}{625} < 0$ function is decreasing at $(2,5)$ as we vary y and hold x fixed

Definitions of partial derivatives of functions of two variables can be extended to cover functions of more than two variables.

If f is a function of n variables $x_1, x_2, \dots, x_n$, then f has n partial derivatives, $f_1(x_1, x_2, \dots, x_n)$, $f_2(x_1, x_2, \dots, x_n)$, $\dots$
 $f_1(x_1, x_2, \dots, x_n)$ $f_2(x_1, x_2, \dots, x_n)$
 $f_n(x_1, x_2, \dots, x_n)$ one with respect to each variable
 $f_{x_n}(x_1, x_2, \dots, x_n)$

Ex $\frac{\partial}{\partial x} \left(\frac{2xy}{1+x^2+y^2} \right) = - \frac{2xy}{(1+x^2+y^2)^2} (x+y)$

Remark!!! If a single variable $f(x)$ has a derivative $f'(a)$ at $x=a$, then f is necessarily continuous at $x=a$.
 This property DOES NOT EXTEND to partial derivative.
 Even if all the first partial derivatives of a function of several variables exist at a point, the function may still fail to be continuous at that point.

Ex $f(x,y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$ (a) Find limit $f(x,y) \rightarrow (x,y)$ approaches $(0,0)$ along the line $y=x$ At $(0,0)$

(b) Prove that f is not continuous at the origin.

(c) Show both $\frac{\partial f}{\partial y}$, $\frac{\partial f}{\partial x}$ exist.

(a) $f(x,y) = 0$ along $y=x$ since $xy \neq 0$. $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 0 = 0$ $y=x$

(b) Since $f(0,0) = 1$, the limit in part (a) shows f is not cont. at $(0,0)$.

(c) $\frac{\partial f}{\partial x} \Big|_{(0,0)} = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = 0$ $\frac{\partial f}{\partial y} \Big|_{(0,0)} = \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k} = 0$